



Artificial Intelligence and the Shaping of Democracy

Webinar 3

–

Algorithms in e-recruiting software

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Webinar 3: „Algorithms in e-recruiting software“

- Everyone goes through an application process, the scenario of this exercise.
- First, you will review applications themselves (resource 1), then use a simulation to see how AI aids recruiters (resource 2).
- Finally, we'll reflect on criteria and biases, examining issues like anti-discrimination and fairness in hiring.

Resource 1: Exercise with applicant ranking

Introduction

We all must go through an application process at some point. That's why we chose the scenario for the following exercise.

The simulation of application processes raises central questions of the modern working world that almost everyone experiences sooner or later.

The aim is to first evaluate applications ourselves and then use a simulation to understand how an AI can proceed to help recruiters in the process. In the final step, we would like to encourage reflection on the criteria and possible biases to critically scrutinize issues such as anti-discrimination and fairness in the world of work.

Steps of the exercise

- In the first step of the exercise, you take the perspective of a recruiter.
- On the following pages, you will see a job advertisement for vocational training in an enterprise and 5 applications for the position.
- Read these and put them in the right order to rank them from 1-5 according to your own criteria to find the right candidate for the open position.
- Perhaps it is easier for you if you print the pages 2-8 to compare them.
- It would be just as practical if you could discuss the ranking and the questions with someone.

Job advertisement: Vocational training as industrial clerks (f/m/d)

What will you learn during your training?

During your training at the German Cleaning Company Samtpfotenrein GmbH, you will gain insights into various departments such as Purchasing, Accounting, Sales and Human Resources. You will acquire knowledge of purchasing products and services, conducting negotiations, controlling expenditure and other commercial processes.

Duration of training

3 years (with the option of shortening to 2.5 years)

Main tasks

- Personnel planning and determination of requirements
- Order processing and warehouse management
- Goods logistics and production control
- Creation of price structures and calculations
- Financial accounting and cost analysis

What should you bring with you?

- Secondary school certificate with solid grades in German and math
- Clear communication and negotiation skills
- Strong interest in business processes and commercial activities
- Confident handling of numbers
- Organizational skills and team spirit
- Willingness to develop professionally
- Careful, reliable way of working and positive demeanor

Your role after the training

After successfully completing your training, you will be responsible for tasks such as creating job advertisements, selecting candidates, coordinating orders and monitoring deliveries. You will also be responsible for the financial management of your work area.



Applications (fictitious persona)

JAMIE SCHMIDT

APPLICATION ID: B001

FIRST NAME: JAMIE LAST NAME: SCHMIDT AGE: 30 EMAIL ADDRESS:
JAMIE.SCHMIDT@EMAIL.COM PHONE NUMBER: 02345 678901 CITY:
SAMPLETOWN STREET: FOREST AVENUE HOUSE NUMBER: 9 POSTAL CODE: 23456

- EDUCATION: SECONDARY SCHOOL CERTIFICATE
- MATHEMATICS GRADE: 2
- GERMAN GRADE: 2
- INTERNSHIP EXPERIENCE: PURCHASING INTERNSHIP DURATION: 2 WEEKS
- DISTANCE TO TRAINING LOCATION: 8 KM
- HOBBIES: ROWING

Application Text:

Hello SAMTPFOTENVEREIN Team,

My name is Jamie and I am interested in the apprenticeship position at your company.

My secondary school diploma demonstrates that I have good skills in German and mathematics. Additionally, I have gained initial experience in a four-week internship in purchasing.

I am reliable and look forward to the challenges of the apprenticeship. I would appreciate a positive response!

Best regards,

Jamie



ROBIN WINTER

APPLICATION ID: B002

FIRST NAME: ROBIN LAST NAME: WINTER AGE: 16 EMAIL ADDRESS:
ROBIN.WINTER@EMAIL.COM PHONE NUMBER: 01234 567890 CITY: SAMPLECITY
STREET: FLOWER PATH HOUSE NUMBER: 15A POSTAL CODE: 12345

- EDUCATION: SECONDARY SCHOOL CERTIFICATE
- MATHEMATICS GRADE: 1
- GERMAN GRADE: 1
- INTERNSHIP EXPERIENCE: SALES INTERNSHIP DURATION: 6 WEEKS
- DISTANCE TO TRAINING LOCATION: 5 KM
- HOBBIES: CHESS

Application Text:

Dear Sir or Madam,

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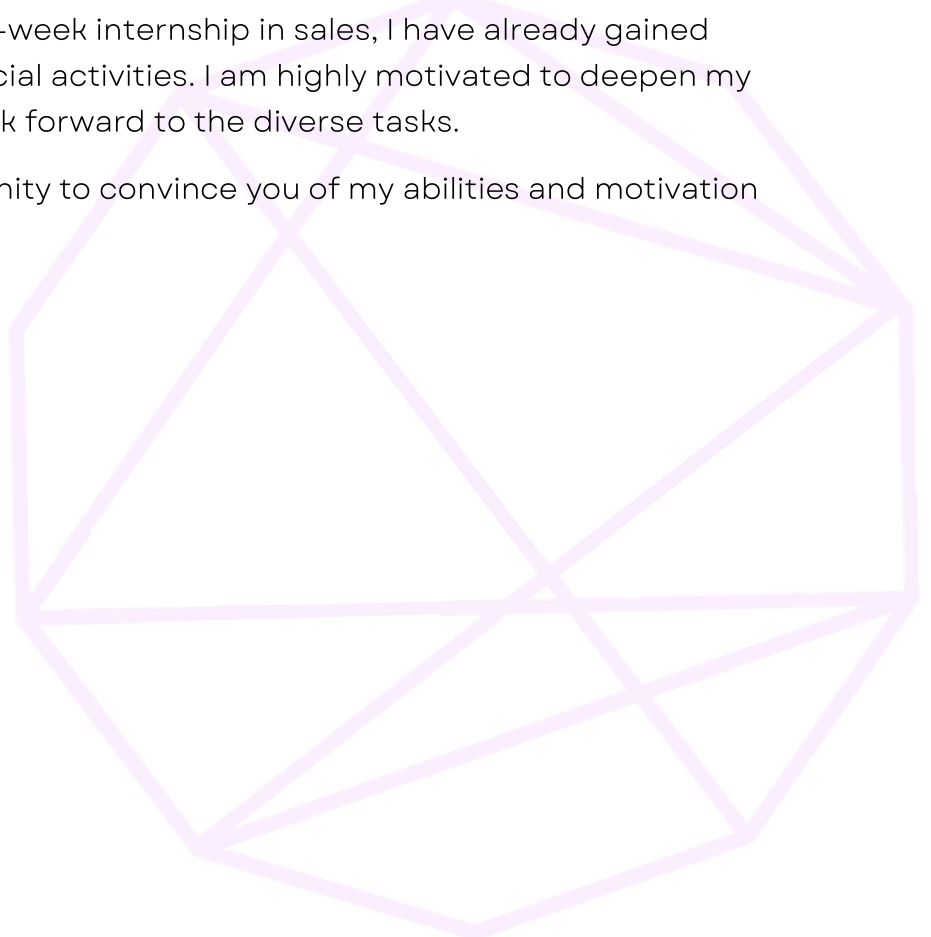
I have read your advertisement with great interest and am convinced that I am the right person for the apprenticeship position as an industrial clerk in your company.

With my secondary school diploma, in which I particularly excelled in German and mathematics, and a six-week internship in sales, I have already gained initial insights into commercial activities. I am highly motivated to deepen my knowledge with you and look forward to the diverse tasks.

Please give me the opportunity to convince you of my abilities and motivation in a personal interview.

Yours sincerely,

Robin Winter





ALEX MÜLLER

APPLICATION ID: B003

FIRST NAME: ALEX LAST NAME: MÜLLER AGE: 15 EMAIL ADDRESS:
ALEX.M@EMAIL.COM PHONE NUMBER: 03456 789012 CITY: SAMPLEQUARTER
STREET: MOUNTAIN STREET HOUSE NUMBER: 22 POSTAL CODE: 34567

- EDUCATION: HIGH SCHOOL DIPLOMA
- MATHEMATICS GRADE: 3 GERMAN GRADE: 3
- INTERNSHIP EXPERIENCE: WAREHOUSE INTERNSHIP DURATION: 2 WEEKS
- DISTANCE TO TRAINING LOCATION: 15 KM
- HOBBIES: GYM

Application Text:

Hi,

I'm Alex and I'm interested in the position.

I was okay in school, especially in mathematics. I also did a short internship.

Thanks and see you soon,

Alex Müller





CASEY WEBER

APPLICATION ID: B005

FIRST NAME: CASEY LAST NAME: WEBER AGE: 16 EMAIL ADDRESS:
CASEY.WEBER@EMAIL.COM PHONE NUMBER: 02345 678912 CITY: OLD TOWN
STREET: GREEN AVENUE HOUSE NUMBER: 12 POSTAL CODE: 67890

- EDUCATION: SECONDARY SCHOOL CERTIFICATE
- MATHEMATICS GRADE: 4
- GERMAN GRADE: 2
- INTERNSHIP EXPERIENCE: NONE INTERNSHIP DURATION: 0 WEEKS
- DISTANCE TO TRAINING LOCATION: 20 KM
- HOBBIES: SLEEPING

Application Text:

Dear Sir or Madam,

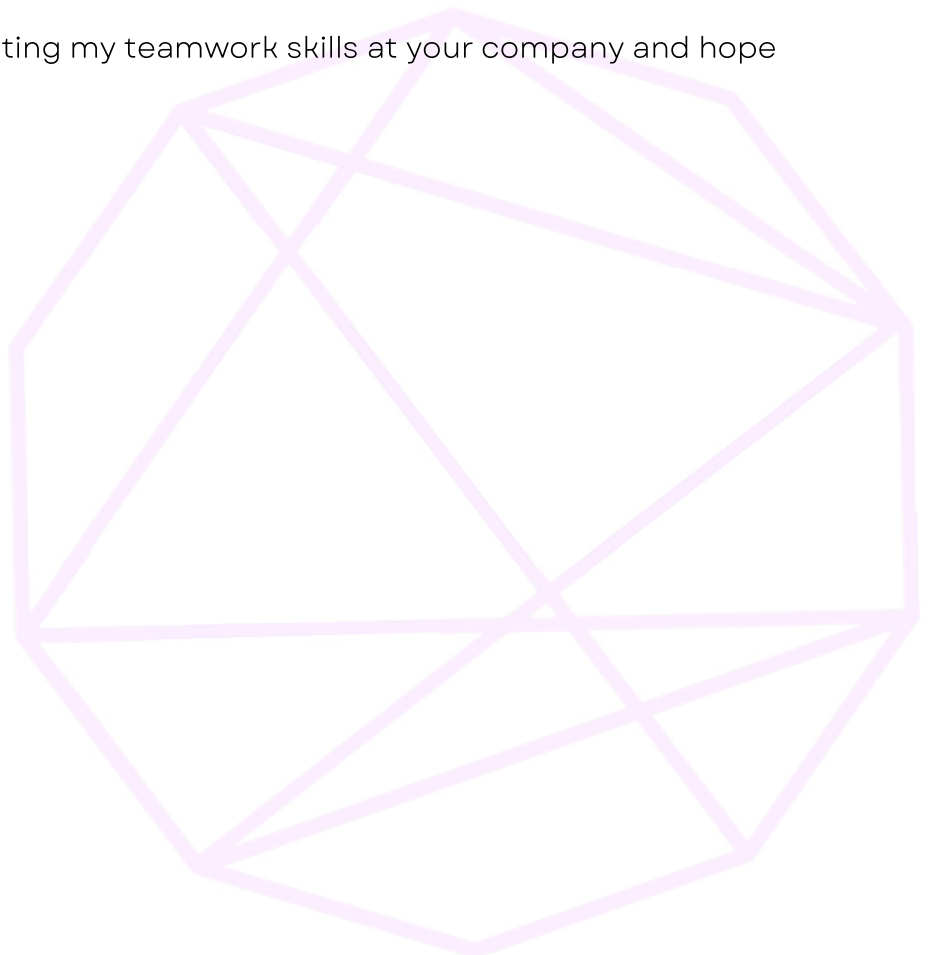
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My name is Casey Weber and I am applying for the apprenticeship position in your company. My strengths are in language subjects, while mathematics is not my main subject. I have not completed any internships yet, but I learn quickly and am particularly good at teamwork.

I look forward to demonstrating my teamwork skills at your company and hope for a positive response.

Yours sincerely,

Casey Weber





SAMANTHA RICHTER

APPLICATION ID: B004

FIRST NAME: SAMANTHA LAST NAME: RICHTER AGE: 19 EMAIL ADDRESS:
SAM.RICHTER@EMAIL.COM PHONE NUMBER: 03456 789123 CITY: NEWCITY
STREET: SUNNY PATH HOUSE NUMBER: 47 POSTAL CODE: 78901

- EDUCATION: SPECIALIZED HIGH SCHOOL
- MATHEMATICS GRADE: 2
- GERMAN GRADE: 2
- INTERNSHIP EXPERIENCE: ACCOUNTING INTERNSHIP DURATION: 8 WEEKS
- DISTANCE TO TRAINING LOCATION: 10 KM
- HOBBIES: RUNNING

Application Text:

Dear Recruitment Team,

My name is Samantha Richter, and I would like to apply for the apprenticeship position in your company.

My specialized high school diploma in business and administration, focusing on business management and accounting, has sparked my passion for commercial activities.

A two-month internship in an accounting department allowed me to apply theoretical knowledge in practice.

I am a goal-oriented and organized person who quickly adapts to new tasks. The opportunity to further develop my skills at your company is very appealing to me.

Thank you for your consideration,

Samantha Richter



Manual applicant ranking

Please rank the candidates from 1-5 according to your own criteria.

Criteria can be, for example

- school-leaving certificate
- internship experience
- confidence in dealing with numbers
- language skills
- interest in commercial processes
- ability to work in a team
- distance to place of work
- hobbies

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You can choose to make a more intuitive assessment or to apply a more detailed points system.

NAME	RANKING
CASEY WEBER	
SAMANTHA RICHTER	
JAMIE SCHMIDT	
ALEX MÜLLER	
ROBIN WINTER	



Manual applicant ranking reflection questions

How did you decide?

What were your motives?

What did you notice?

How to continue?

In the second exercise (Resource 2), where we will use a simulation to understand how an AI can proceed to help recruiters in the process.

In the end, we would like to encourage reflection on the criteria and possible biases to critically scrutinize issues such as anti-discrimination and fairness in the world of work.

Go to resource 2 and start it right away.





Resource 2: AI-based ranking - simulation

After completing the manual application ranking, you can now take on the role of the applicant and the recruiter with a simulation.

Introduction

- Go to the following website: <https://iug.htw-berlin.de/workshop/HR2/>
- You can change the language at the top right.
- Then follow the instructions below





Instructions

From the applicant's point of view

- First, let's take the role of the applicant.
- On the landing page of the simulator, choose the role applicant.
- On the applications from the manual ranking, there is an ID for each application. These can now be entered into the form and submitted as examples. You will only have to manually fill in the drop-down fields for certain criteria. After submitting you will see the place on the ranking page. You can use the bias button to change the ranking and see how this affects the position. Each time the button is pressed, a different bias is included and the button changes. However, you will only see the ranking of the entered application, not that of the other applications. Identifying the prejudices is done in the recruiters' view later.
- Experiment now with the application form: By pressing the implemented "Reload" button, the application will be deleted, and you can repeat the process with changed data. You are also given the opportunity to create your own application. You need to choose a fictitious applicant ID (BID), for example 'Test', and are asked not to use the IDs B001 to B005. This self-created application can either be particularly either particularly convincing or deliberately inadequate, to differentiate between application styles in the ranking. Of course, you only see how you were ranked.
- Experimenting with the ranking enables one to gain a deeper understanding of how certain application details can influence the result.

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Reflection questions as applicant

How does it feel to know a computer, not a person, will review your application?

How could you design your application to influence the ranking?



Role change: now we take on the perspective of a recruiter

- Let us now take over the role of the recruiter.
- Select the role as a recruiter on the landing page. After repeating the job advertisement, you will see the ranking with the list of the 5 applications that you already know from the manual ranking. The individual scores in each criterion are incorporated into an overall score, which is referred to as the “AI score”. Now find the bias that is not always directly recognizable in the listed criteria.
- By pressing the bias button, you can play through various biases and observe their effects on the ranking.
- In the unchanged state without bias, the “No bias” button is displayed.
- Analyze the ranking and identify the bias.

Reflection questions as a recruiter

What are the results of the ranking?

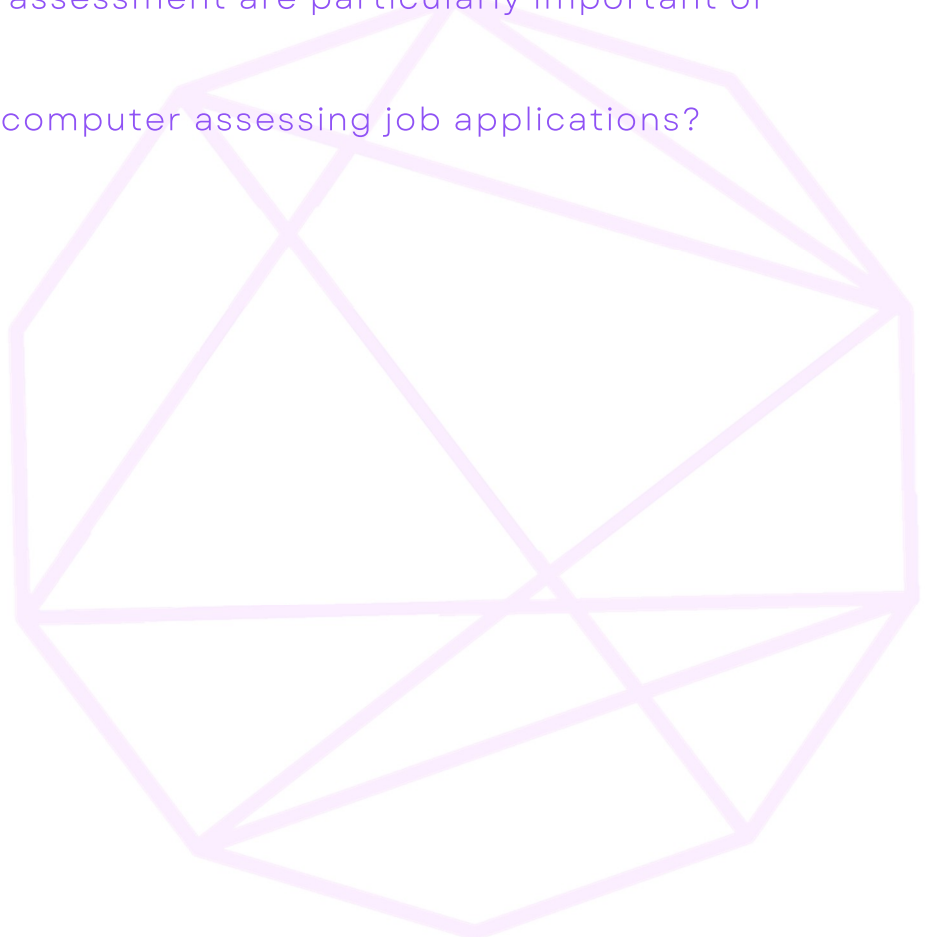
Were there any surprises?

How did you find the task of identifying the bias?

Was it easy or challenging?

What aspects of the AI assessment are particularly important or worrying to you?

How do you feel about computer assessing job applications?





Revealing the bias

- 1st bias: School-leaving certificate. “HIGH SCHOOL DIPLOMA” gets a lot of points.
- 2nd bias: Age. Points are deducted for applicants over 20 years old.
- 3rd bias: Team sport. Practicing a team sport such as aft rowing leads to additional points for teamwork.

Bias or criterion?

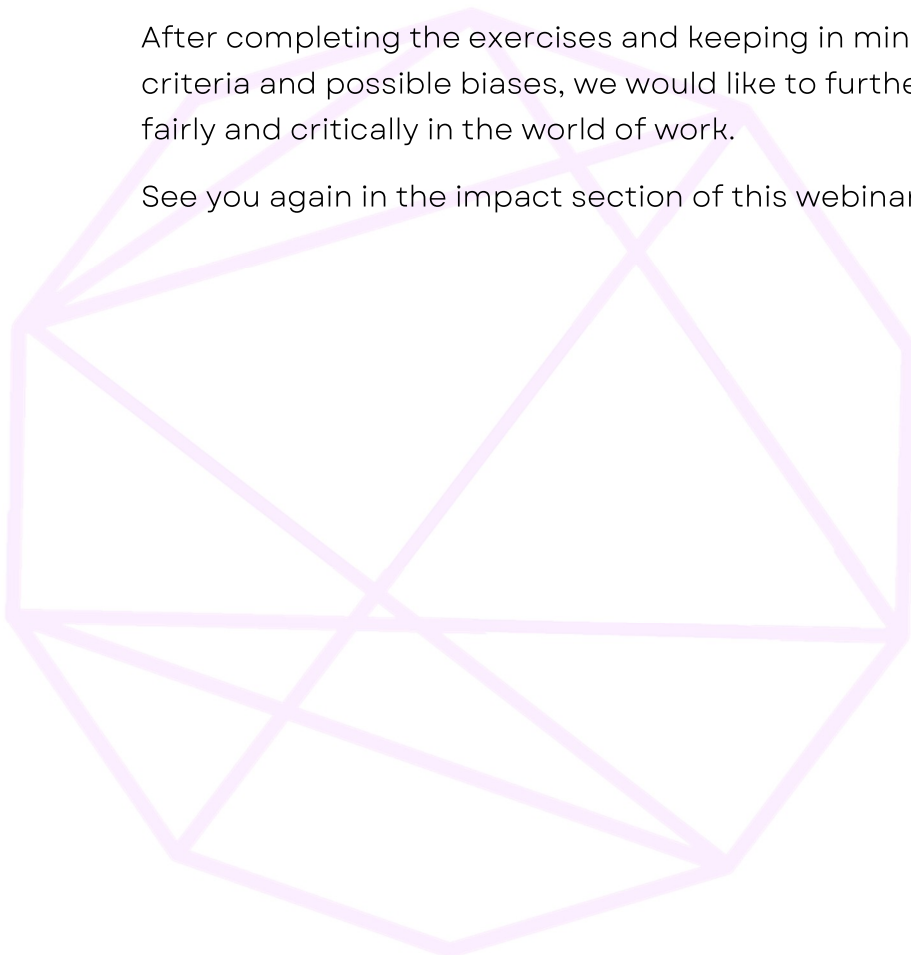
- Awarding points for the high school diploma could be biased, as the school-leaving qualification is given excessive weight although the job advertisement only requires an intermediate school leaving certificate. In this case, it could be unfair to rate applicants with an intermediate school-leaving certificate, even if they fulfill all other criteria.
- A lower rating due to age is clearly a prejudice. It is not based on objective criteria, but on assumptions about the performance and adaptability of older people.
- If team sports are only one of several possible pieces of evidence of team skills (which is the case here), they can be considered as one criterion.

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What's next?

After completing the exercises and keeping in mind the difference between criteria and possible biases, we would like to further reflect on how to use AI fairly and critically in the world of work.

See you again in the impact section of this webinar.





Literature

Bundesarbeitskreis Arbeit und Leben/ HTW Forschungsgruppe „Informatik und Gesellschaft“: KILE. Available at <https://kile.arbeitundleben.de>.



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